

Lecture 3

Boolean Algebra

Postulates and duality · basic theorems · DeMorgan's theorem ·
Boolean functions · algebraic simplification

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 2, §2.1–2.5



$$(xy)' = x' + y'$$

Today's roadmap

1

§2.1–2.3

Postulates

The rules we assume

2

§2.4

Theorems

Proved from the postulates

3

§2.4

DeMorgan

Complement a whole expression

4

§2.5

Functions

Tables, expressions, gates

By the end of class you can...

- State the postulates and apply the duality principle
- Prove theorems one postulate per step
- Complement expressions with DeMorgan's theorem
- Evaluate expressions with the right precedence
- Simplify expressions and count literals

Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	Homework 1 collected
0:05 – 0:20	Postulates and duality	—
0:20 – 0:45	Basic theorems and proofs	Examples 1, 2 · Your turn 1
0:45 – 0:55	Break	
0:55 – 1:15	DeMorgan’s theorem, precedence	Example 3
1:15 – 1:40	Boolean functions and simplification	Examples 4, 5 · Your turn 2
1:40 – 1:50	Concept check and wrap-up	Homework preview

The postulates of Boolean algebra

§2.1–2.3

Postulate	(a)	(b)
Closure	$x + y \in B$	$x \cdot y \in B$
Identity	$x + 0 = x$	$x \cdot 1 = x$
Commutative	$x + y = y + x$	$xy = yx$
Distributive	$x(y + z) = xy + xz$	$x + yz = (x + y)(x + z)$
Complement	$x + x' = 1$	$x \cdot x' = 0$
Elements	at least two: $0 \neq 1$	

Duality

Swap $+$ with $\cdot$ and 0 with 1 : every true identity stays true.

So each postulate and theorem comes in an (a) and a (b) form.

New to you

$x + yz = (x + y)(x + z)$ has no counterpart in ordinary algebra.

The basic theorems

§2.4

Theorem	(a)	(b)
1 Idempotent	$x + x = x$	$x \cdot x = x$
2 Null element	$x + 1 = 1$	$x \cdot 0 = 0$
3 Involution	$(x')' = x$	
4 Associative	$x + (y + z) = (x + y) + z$	$x(yz) = (xy)z$
5 DeMorgan	$(x + y)' = x'y'$	$(xy)' = x' + y'$
6 Absorption	$x + xy = x$	$x(x + y) = x$
Consensus	$xy + x'z + yz = xy + x'z$	(dual)

Operator precedence

1. Parentheses
2. NOT
3. AND
4. OR

$A + B'C$ means $A + ((B')C)$.

Example 1 — Proving a theorem

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EXAMPLE

Prove $x + x = x$ and $x + xy = x$ from the postulates, one postulate per step.

Step	Justification
$x + x = (x + x) \cdot 1$	identity 2(b)
$= (x + x)(x + x')$	complement 5(a)
$= x + xx'$	distributive 4(b)
$= x + 0$	complement 5(b)
$= x$	identity 2(a)

Every line is one rule. Name the rule, and the grader can follow you.

- 1 Absorption: $x + xy = x \cdot 1 + xy$ (identity)
- 2 $= x(1 + y)$ (distributive)
- 3 $= x \cdot 1$ (null element, Theorem 2)
- 4 $= x$ (identity)
- 5 Dual for free: $x(x + y) = x$

Example 2 — Simplify algebraically

§2.4

EXAMPLE

Simplify (a) $x(x' + y)$, (b) $x + x'y$, (c) $(x + y)(x + y')$.

Strategy

Look for a variable and its complement in the same place:
 $x + x' = 1$ or $xx' = 0$.

Distribute to bring them together.

(b) is the dual of (a), so its answer is the dual of (a)'s.

Check any result by substituting values.

1 (a) $x(x' + y) = xx' + xy$ (distributive)

2 $= 0 + xy$ (complement)

3 $= xy$ (identity)

4 (b) $x + x'y = (x + x')(x + y)$ (distributive, 4b)

5 $= 1 \cdot (x + y)$ (complement)

6 $= x + y$ (identity)

7 (c) $(x + y)(x + y') = x + yy'$ (distributive, 4b)

8 $= x + 0$ (complement)

9 $= x$ (identity)

Your turn 1 — Simplify and prove

§2.4

YOUR TURN

≈ 10 min

- (a) Simplify $xyz + x'y + xyz'$ to one literal.
- (b) Show that $(A + B)(A + C) = A + BC$ by multiplying out.
- (c) Prove $(x + y)' + (x + y')' = x'$.

Name the postulate or theorem you use at each step.

Your turn 1 — Solution

§2.4

SOLUTION

1 (a) Reorder: $xyz + xyz' + x'y$ (commutative)

2 $= xy(z + z') + x'y$ (distributive)

3 $= xy \cdot 1 + x'y = xy + x'y$ (complement, identity)

4 $= y(x + x')$ (distributive)

5 $= y \cdot 1 = y$ (complement, identity)

6 (b) Multiply out: $AA + AC + AB + BC$ (distributive)

7 $= A + AC + AB + BC$ (idempotent: $AA = A$)

8 $= A(1 + C + B) + BC$ (distributive)

9 $= A \cdot 1 + BC$ (null element: $1 + x = 1$)

10 $= A + BC$ (identity)

11 (c) $(x + y)' + (x + y')' = x'y' + x'y$ (DeMorgan)

12 $= x'(y' + y)$ (distributive)

13 $= x' \cdot 1$ (complement)

14 $= x'$ (identity)

Break

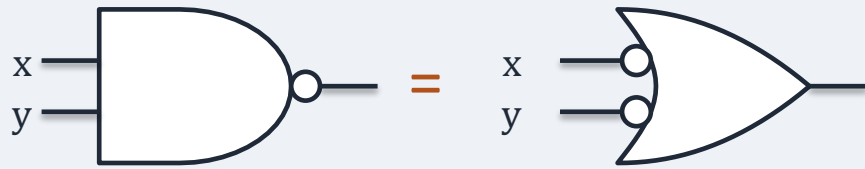
10 minutes — we resume at the 0:55 mark

Next: DeMorgan's theorem and Boolean functions

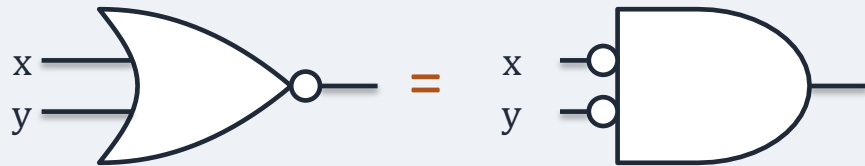
DeMorgan's theorem

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$(xy)' = x' + y'$: NAND = OR with inverted inputs



$(x + y)' = x'y'$: NOR = AND with inverted inputs



x	y	x+y	(x+y)'	x'y'
0	0	0	1	1
0	1	1	0	0
1	0	1	0	0
1	1	1	0	0

Complement of an expression

Take the dual, then complement every literal.

$(A + B + C)' = A'B'C'$; it extends to any number of variables.

Example 3 — Complementing functions

§2.4

EXAMPLE

Find F' for (a) $F = AB + C'D$ and (b) $F = (A + B + C')(A' + B')C$.

Two methods

1. DeMorgan outside in, one level at a time.
2. Dual of F , then complement each literal.

Check (b) at $A = 1, B = 0, C = 1$:

$F = (1)(1)(1) = 1$, so F' must be 0.

$F' = 0 + 0 + 0 = 0$ ✓

- 1 (a) DeMorgan on the OR: $F' = (AB)'(C'D)'$
- 2 DeMorgan on each AND: $(A' + B')(C'' + D')$
- 3 Involution $C'' = C$: $F' = (A' + B')(C + D')$
- 4 (b) DeMorgan on the AND: $(A + B + C')' + (A' + B')' + C'$
- 5 DeMorgan inside: $(A + B + C')' = A'B'C$
- 6 and $(A' + B')' = A''B'' = AB$
- 7
$$F' = A'B'C + AB + C'$$
- 8 Method 2: dual $ABC' + A'B' + C \rightarrow$ complement literals ✓

Precedence decides the function

§2.4

Evaluate $F = A + B'C + D$

Order: parentheses, NOT, AND, OR.

At $A = 0, B = 1, C = 1, D = 0$:

$B' = 0, B'C = 0, F = 0 + 0 + 0 = 0$

The trap

$A + B'C + D$ is not $(A + B')(C + D)$.

They agree above by coincidence; the table shows a row where they differ.

ABCD	$A+B'C+D$	$(A+B')(C+D)$	Agree?
0110	0	0	yes
1100	1	0	no
1000	1	0	no
1010	1	1	yes

When you mean OR-before-AND, write the parentheses. Verilog uses the same order: NOT, then AND, then OR.

Boolean functions

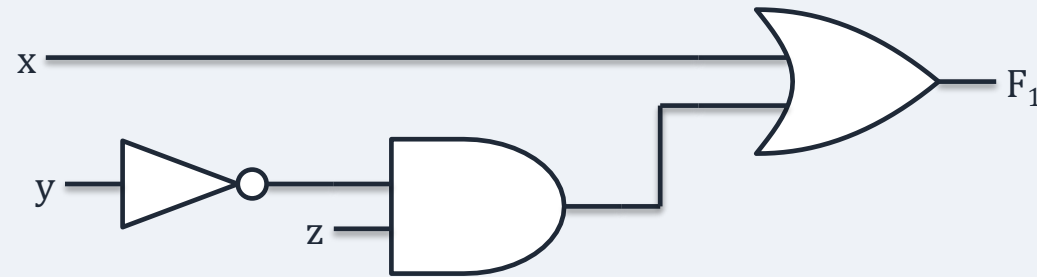
§2.5

x	y	z	F ₁	F ₂
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	0
1	1	1	1	0

$$F_1 = x + y'z$$

$$F_2 = x'y'z + x'yz + xy'$$

Gate diagram of $F_1 = x + y'z$



One table, many expressions

A function has exactly one truth table but many equivalent expressions. The expression you pick decides the gates you build.

Literal: a variable or its complement, e.g. x , y' .

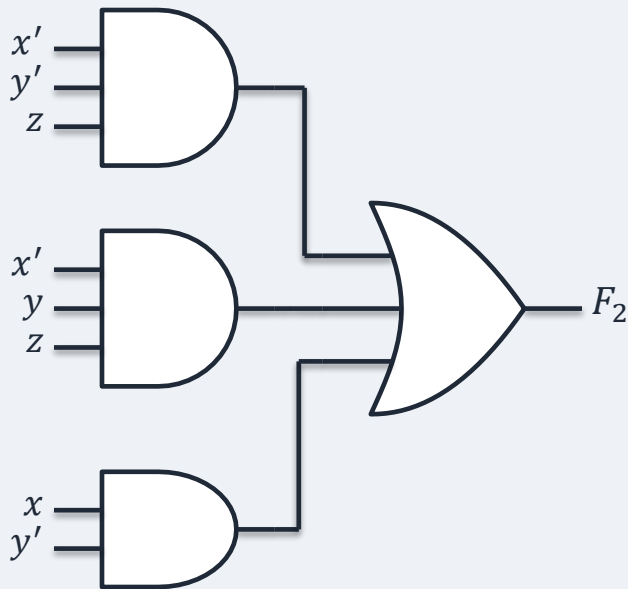
Example 4 — Fewer literals and gates

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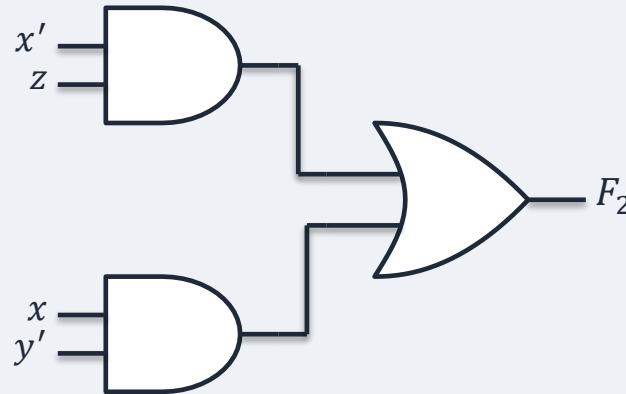
EXAMPLE

Simplify $F_2 = x'y'z + x'yz + xy'$ and compare the two circuits.

As given: 8 literals



Simplified: 4 literals



- 1 Find two terms that differ in one variable:
 $x'y'z, x'yz$
- 2 Factor: $x'z(y' + y)$ (distributive)
- 3 $= x'z \cdot 1 = x'z$ (complement, identity)
- 4 $F_2 = x'z + xy'$
- 5 Both are $\Sigma m(1,3,4,5)$ ✓
- 6 $8 \rightarrow 4$ literals, $3 \rightarrow 2$ AND gates.

Example 5 — Consensus and absorption

§2.4

EXAMPLE

Simplify $F = AB + A'C + BC + AB'D + A'BD$.

Consensus theorem

$$xy + x'z + yz = xy + x'z$$

yz is covered: when $y = z = 1$, either xy or $x'z$ is 1.

Proof: $yz = xyz + x'yz$, absorbed by xy and $x'z$.

Useful form: $x + x'y = x + y$

- 1 AB has A , $A'C$ has A' : the consensus is $B \cdot C$
- 2 So BC is covered: drop it (consensus)
- 3 $AB + AB'D = A(B + B'D)$ (distributive)
- 4 $= A(B + D) = AB + AD$ ($x + x'y = x + y$)
- 5 $AB + A'BD = B(A + A'D)$ (reuse AB : $x = x + x$)
- 6 $= B(A + D) = AB + BD$ ($x + x'y = x + y$)
- 7 Collect: $F = AB + A'C + AD + BD$ (8 literals, from 13)
- 8 Check: both are $\Sigma m(2,3,5,6,7,9,11..15)$ ✓

Your turn 2 — Functions

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YOUR TURN

≈ 10 min

- (a) Find the complement of $F = x(y'z' + yz)$.
- (b) Simplify $F = ABC + ABC' + A'B$ to one literal.
- (c) Simplify $F = xy + x'z + yz$ and draw the two-level circuit.

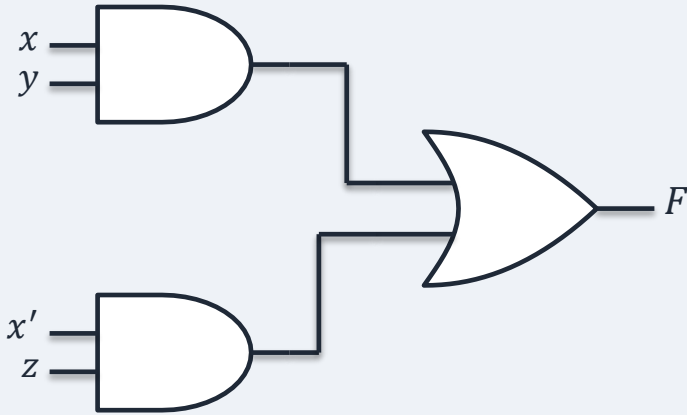
Count the literals before and after each simplification.

Your turn 2 — Solution

§2.5

SOLUTION

(c) $F = xy + x'z$



1 (a) $F' = x' + (y'z' + yz)'$ (DeMorgan)

2 $= x' + (y'z')'(yz)'$ (DeMorgan)

3 $= x' + (y + z)(y' + z')$ (DeMorgan, involution)

4 (b) $ABC + ABC' = AB(C + C') = AB$ (distributive, complement)

5 $AB + A'B = B(A + A') = B$ (distributive, complement)

6 (c) xy has x , $x'z$ has x' : consensus yz

7 Drop yz : $F = xy + x'z$ (consensus)

8 Literals: $6 \rightarrow 4$

Concept check

1

What is the dual of $x + yz = (x + y)(x + z)$?

2

Why must the complement of $AB + C'D$ keep its parentheses?

3

Does $x + x = 2x$ in Boolean algebra? Why not?

4

When is a term the consensus of two others?

Key takeaways

Duality

$$+ \leftrightarrow \cdot \quad 0 \leftrightarrow 1$$

Distributive (b)

$$x + yz = (x + y)(x + z)$$

Absorption

$$x + xy = x$$

DeMorgan

$$(x + y)' = x'y'$$

Useful

$$x + x'y = x + y$$

Consensus

$$xy + x'z + yz = xy + x'z$$

Every simplification in this course is these rules applied in order.

Homework #2 due at the start of Lecture 5

Next: canonical and standard forms, other gates